Prepare your laptop for the labs

One script, once, about 10 minutes

What you need

- A laptop with Linux (Ubuntu, Debian, Mint, Pop!_OS, Fedora, Arch, Manjaro or openSUSE).
- Internet and your password (the script asks for it once, to install packages).
- The file os2026-setup.sh from the course channel, saved in your Downloads folder.
- No Linux on your laptop? Tell the assistant before the lab: a Linux virtual machine or a live USB works too.

Three steps

1. Open a terminal (Ctrl+Alt+T on most systems) and go where the file is:

```
cd ~/Downloads        # or ~/Descarcari, wherever you saved it
```

2. Run it and wait. It installs the RISC-V compiler and QEMU, downloads xv6 into ~/xv6-riscv, builds it and boots it once to test it:

```
bash os2026-setup.sh
```

3. The last line must be green: "All set." From now on you start xv6 with:

```
cd ~/xv6-riscv && make qemu
# leave xv6: Ctrl-A, release, then X
```

On Ubuntu 22.04, Mint 21 and Debian 11 the script also builds a newer QEMU (the one they ship is too old for xv6): it takes 5 to 10 minutes more, let it run.

Safe to repeat

Run the script again whenever something looks wrong. It never deletes your work in ~/xv6-riscv. To only test, without installing anything: bash os2026-setup.sh --check

If something goes wrong

You see	Do this
Unknown distribution	Show it to the assistant: we install the four tools by hand (git, make, a riscv64 gcc, qemu-system-riscv64).
sudo: password, or "not in the sudoers file"	Type your login password (nothing is shown while you type). Without admin rights, ask the assistant.
"Could not get lock" (apt)	The system is updating in the background. Wait two minutes and run the script again.
"Failed to get write lock" (QEMU)	Another xv6 is still running. Close it (Ctrl-A, X) or run pkill qemu-system-riscv64 , then retry.
git clone failed	Check the internet connection (try opening github.com), then run the script again.